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Machine learning-based integration of pericoronary adipose tissue and clinical risk factors for cardiovascular risk prediction in type 2 diabetes: a retrospective cohort study

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Abstract

Background Cardiovascular disease remains the predominant cause of morbidity and mortality in individuals with type 2 diabetes mellitus (T2DM). Traditional risk models are limited in predictive accuracy. Pericoronary adipose tissue (PCAT), a novel imaging biomarker of vascular inflammation, may offer additional prognostic value. Therefore, this study aimed to develop and validate a machine learning model that integrates PCAT parameters with clinical risk factors to improve the accuracy of cardiovascular risk prediction in individuals with T2DM.

Methods This study retrospectively enrolled 686 hospitalized T2DM patients from four branches of Guangdong Provincial Hospital of Chinese Medicine between January 2017 and December 2021. PCAT-FAI and volume index were measured using coronary CTA. Major adverse cardiovascular events (MACE) were recorded during follow-up. Eight machine learning algorithms were applied, and multiple evaluation metrics were used to compare the predictive performance of the models. Feature contributions in the best-performing model were interpreted using both feature importance ranking and SHapley Additive exPlanations (SHAP) values.

Results A total of 183 patients experienced MACE during the mean 38.4 months of follow-up. Among the eight machine learning models evaluated, the XGBoost model performed the best in predicting MACE in patients with T2DM. In the internal validation of the training set, the AUC was 0.818 (95% *CI* 0.777—0.858), and in the external test set, the AUC was 0.809 (95% *CI* 0.700—0.918). Additionally, the XGBoost model outperforms other models in all evaluation metrics (accuracy = 0.824, specificity = 0.882, F1 score = 0.654, Brier score = 0.248). In the feature importance analysis of the prediction model, RCA-FAI in the PCAT parameters consistently ranked among the top three in eight ML models. Further SHAP analysis indicated that RCA-FAI, body mass index (BMI), and the monocyte/high-density lipoprotein cholesterol ratio (MHR) were the most influential factors for MACE in patients with T2DM.

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Conclusion This study demonstrates the independent predictive value of PCAT parameters for long-term cardiovascular risk in patients with T2DM. The XGBoost model showed promise as a potential clinical decision support tool. Integrating PCAT parameters with conventional risk factors may improve the identification of high-risk individuals and enhance the ability to predict MACE in this population.

Clinical trial registration ChiCTR2400079869

Keywords Type 2 diabetes mellitus, Machine learning, Pericoronary adipose tissue, Major adverse cardiovascular events, Predictive model

Introduction

Type 2 diabetes mellitus (T2DM) is an escalating global public health challenge, with a steadily increasing prevalence worldwide. According to recent surveillance data, an estimated 589 million adults worldwide were living with T2DM in 2024, a figure projected to rise to 853 million by 2050 [1]. T2DM is characterized by chronic hyperglycemia and insulin resistance, which promote metabolic dysfunction and significantly increase the risk of cardiovascular disease (CVD) and adverse clinical outcomes [2]. According to the American Heart Association (AHA), the majority of deaths among individuals with T2DM are attributable to CVD. Furthermore, the one-year mortality rate following acute cardiovascular events is approximately 15% in this population, nearly threefold higher than that observed in the general population [3]. This further underscores the urgent need to accurately predict cardiovascular risk in individuals with T2DM.

Currently, widely used CVD risk prediction models (e.g., Framingham Risk Score, SCORE system) are primarily developed based on traditional risk factors, including age, blood pressure, and metabolic indicators [4, 5]. However, these conventional tools exhibit significant limitations when applied to T2DM patients. T2DM patients exhibit unique cardiovascular pathologies, including chronic coronary inflammation, endothelial dysfunction, enlarged necrotic lipid cores, and attenuated fibrous caps [6, 7]. Conventional risk models fail to adequately incorporate these diabetes-specific pathological processes, resulting in systematic biases and underestimation of cardiovascular risk in this population. A multicenter study demonstrated that approximately 17% of T2DM patients could not be accurately stratified into established cardiovascular risk categories using conventional scoring systems, thereby compromising the timeliness and effectiveness of early clinical interventions [8]. Consequently, recent studies have increasingly explored the integration of imaging-based indicators and biological markers to improve cardiovascular risk prediction in T2DM. Nevertheless, the overall enhancement in predictive performance remains suboptimal, highlighting the need for more advanced and interpretable modeling approaches.

Pericoronary adipose tissue (PCAT), a specialized component of epicardial fat that surrounds the coronary arteries, plays a pivotal role in the development of cardiovascular disease (CVD). Through paracrine signaling, PCAT releases various pro-inflammatory cytokines and adipokines, including adiponectin, which engage in dynamic cross-communication with adjacent vascular structures [9]. These bioactive mediators can induce endothelial dysfunction, promote monocyte adhesion and transmigration, and stimulate smooth muscle proliferation and extracellular matrix remodeling, thereby directly driving the progression of atherosclerosis [10]. Prior studies have demonstrated that elevated PCAT-FAI is significantly associated with plaque vulnerability and a heightened risk of major adverse cardiovascular events (MACE) [11, 12]. In a cohort of 503 patients with suspected coronary artery disease (CAD), each 10 HU increase in PCAT-FAI was linked to a 15–20% rise in MACE risk [13]. However, current studies have largely focused on short-term outcomes or non-diabetic cohorts [14, 15]. The long-term prognostic value of PCAT parameters in individuals with T2DM and their potential interactions with conventional risk factors remain insufficiently explored.

Recently, machine learning (ML) has shown potential to significantly improve cardiovascular risk prediction by capturing complex, nonlinear relationships between variables [16, 17]. Compared to traditional regression models, ML algorithms can simultaneously integrate imaging biomarkers with multidimensional clinical data, while leveraging feature engineering techniques to explore inter-variable interactions. Previous imaging-based studies have demonstrated that ML-driven models can effectively stratify CAD patients into distinct risk categories [18]. However, systematic comparisons of the predictive performance of different ML algorithms for MACE in T2DM patients are lacking, and the longitudinal stability of these models remains unvalidated. Therefore, this study aimed to develop and validate ML-based predictive models that integrate PCAT parameters with clinical risk factors. It further sought to systematically assess the performance and robustness of various ML algorithms in a T2DM

cohort, with the goal of enhancing the accuracy of cardiovascular risk prediction.

Materials and methods

Study population

This study retrospectively analyzed 1,403 patients with T2DM, aged 18 to 80 years, who underwent CCTA at Guangdong Provincial Hospital of Chinese Medicine between January 2017 and December 2021. The diagnosis of T2DM was based on the American Diabetes Association (ADA) criteria and confirmed through medical records [19]. The exclusion criteria were as follows: (1) prior coronary revascularization (PCI or CABG), (2) congenital coronary anomalies, (3) malignant tumor, poor-quality CCTA images, (4) missing relevant data or missing follow-up. After applying these criteria, a total of 686 eligible patients were included. Among them, 584 patients from the Dade Road hospital (DDRH), Fangcun branch (FCB), and University Town branch (UTB) were used as the training cohort, while 102 patients from the Ersha Island branch (ESIB) served as the external testing cohort. Among the entire cohort, 183 patients experienced MACE during follow-up, whereas 503 did not (Fig. 1). The study was approved by the Ethics Committee of Guangdong Provincial Hospital of Traditional Chinese Medicine (No.: BE2022-173–01). Registered at https://www.chictr.org.cn/, registration number: ChiCTR2400079869.

Follow-up

Comprehensive follow-up data were obtained via medical record review, telephone interviews, and outpatient visits. The primary endpoint was the occurrence of MACE, defined as coronary revascularization, hospitalization for heart failure, hospitalization for acute coronary syndrome, or cardiac death. To ensure the validity of the study findings, all included patients underwent CCTA at baseline. Patients who experienced MACE within 90 days of baseline CCTA were excluded to minimize reverse causality, consistent with prior cardiovascular imaging studies [20, 21]. The 90-day interval was selected to provide a balance between excluding events likely attributable to concurrent disease activity and retaining sufficient patient numbers for robust analysis.

CCTA image acquisition and analysis

All CCTA examinations were conducted using a dual-source CT scanner (SOMATOM Definition Flash, Siemens Healthineers). The scanning protocol included prospective ECG gating, with tube voltage automatically adjusted between 100 and 120 kV based on the patient's body mass index (BMI). Iopromide (370 mg I/mL) was administered at a rate of 5 mL/s, and images were acquired with a slice thickness of 0.6 mm. The scan range extended from the carinal level to 2 cm below the diaphragm, ensuring complete visualization of the coronary vasculature. After image acquisition, CCTA data were automatically transferred to a dedicated post-processing workstation (Shukun Technology, China) for offline analysis. Coronary artery stenosis was assessed using the Coronary Artery Disease Reporting and Data System (CAD-RADS) [22], with stenoses of 50% or greater classified as significant stenoses. For accuracy and consistency, two radiologists independently performed all measurements without knowledge of clinical outcomes, and any discrepancies were resolved by a senior radiologist.

PCAT processing and analysis

PCAT analysis was conducted for the right coronary artery (RCA), left anterior descending artery (LAD), and left circumflex artery (LCX). Quantification of the FAI and volume index was automatically performed using the FAI analysis software (1.2.1, Shukun Technology, China). The analysis range was defined as 10 to 50 mm distal from the RCA origin and from the vessel origin to 40 mm for both the LAD and LCX (Fig. 2). FAI was calculated as the average mean CT attenuation value of adipose tissue located within a radial distance from the outer vessel wall equivalent to the vessel diameter, using a predefined attenuation threshold of −190 to −30 HU [23].

Feature collection and data preprocessing

Cardiovascular risk factors relevant to T2DM patients were collected through an extensive literature review and expert consultation [24, 25], encompassing demographic information, detailed medical history, laboratory parameters, medication history, and imaging data—34 variables in total. Definitions and classification criteria for clinical history variables are provided in Supplementary Method 1. To ensure the accuracy of laboratory measurements, all participants fasted for at least 8 h prior to enrollment. Venous blood samples were collected to assess triglycerides (TG), total cholesterol (TC), low-density lipoprotein cholesterol (LDL-C), and high-density lipoprotein cholesterol (HDL-C), monocyte count (MONO), glycated hemoglobin (HbA1c) and estimated glomerular filtration rate (eGFR). The monocyte-to-HDL ratio (MHR) was calculated as MONO divided by HDL-C.

Variables with > 30% missingness were excluded. Missing values in the remaining variables were handled using multiple imputation by chained equations (MICE; m = 5) implemented in the R package mice (predictive mean matching for continuous variables and logistic regression for categorical variables). To avoid information leakage, imputations were performed separately for the training and external cohorts and, within each cross-validation

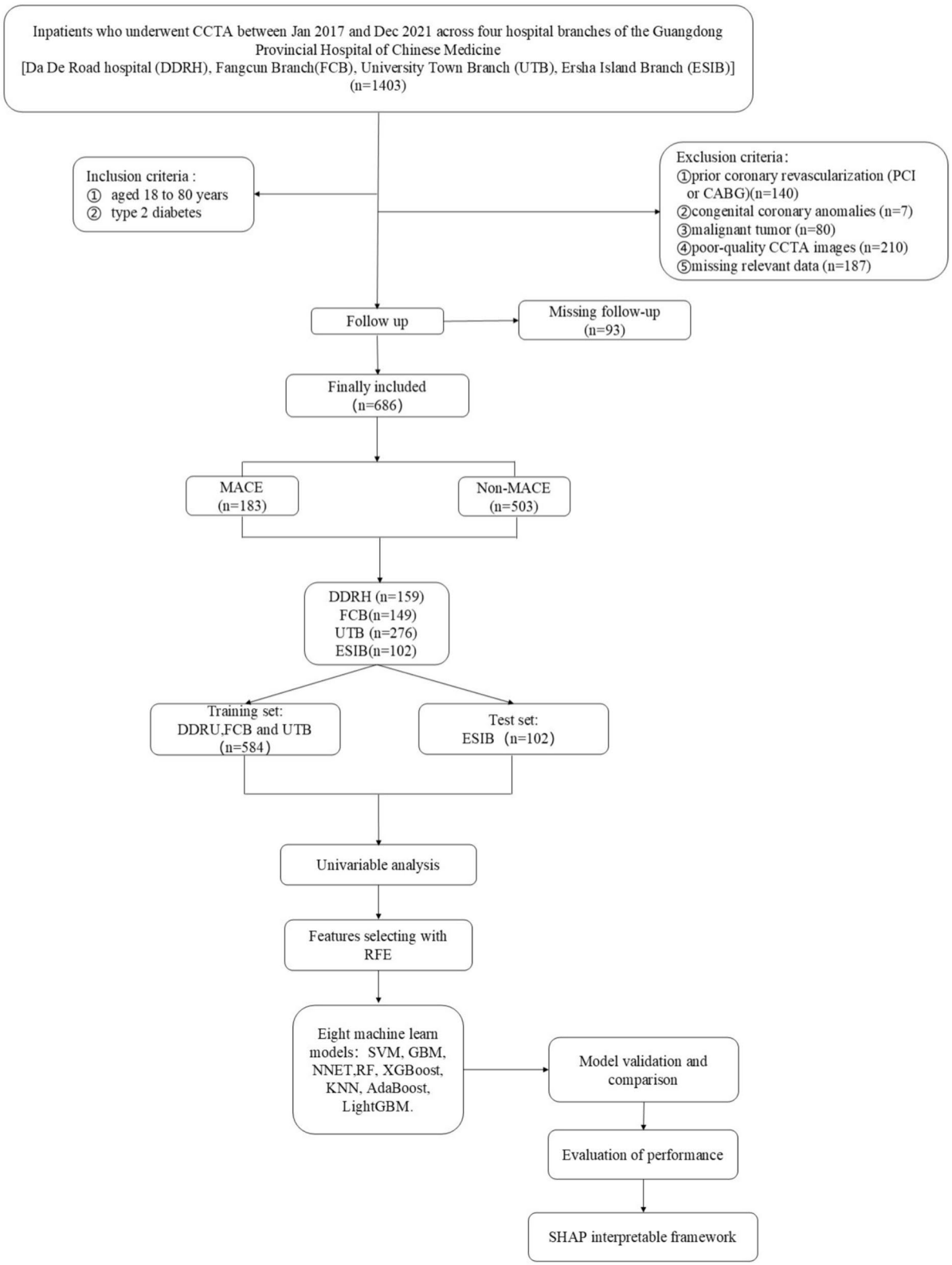


Fig. 1 Flow diagram of the study cohort

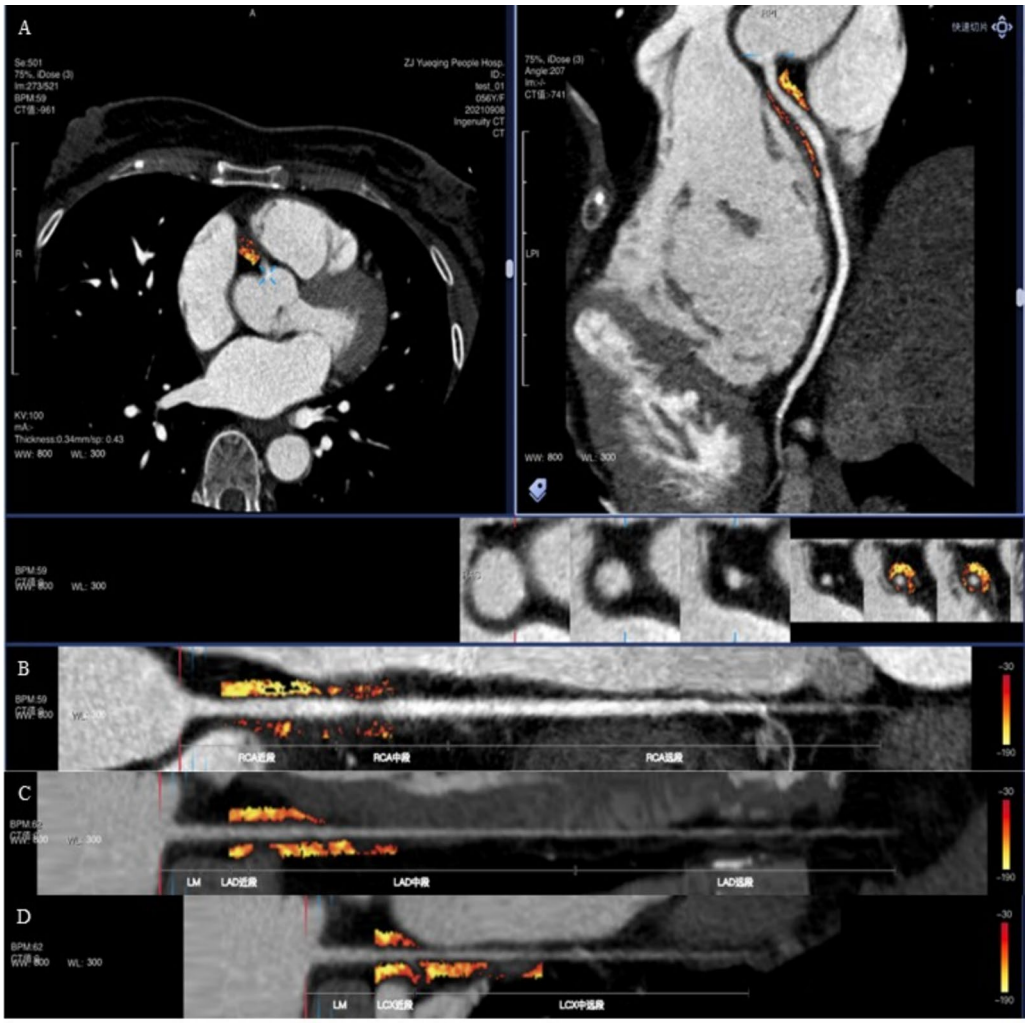


Fig. 2 Schematic diagram of PCAT-FAI measured by CCTA in T2DM patients. **A** Mediastinum window, **B** cross-sectional measurement of RCA-FAI, **C** cross-sectional measurement of LAD-FAI, **D** cross-sectional measurement of LCX-FAI

split, independently for the training and validation subsets. Estimates from the five completed datasets were pooled using Rubin's rules. Multiple imputation was used for remaining incomplete data. Continuous variables were standardized using Z-score normalization, and categorical variables were transformed via one-hot encoding.

Feature selection strategy

Before model development, candidate variables were selected based on clinical experience and previous studies. To reduce dimensionality and potential multicollinearity, variables associated with MACE at $P < 0.05$ in univariate analysis were initially retained as candidate predictors. This step was followed by recursive feature elimination (RFE) based on SVM. Recognizing that univariate filtering may exclude variables that contribute through interactions, we performed sensitivity analyses using XGBoost's embedded feature importance measures (gain and SHAP values). These confirmed the stability of the final predictor set, as the same key variables consistently ranked among the top contributors in the XGBoost model. Accordingly, the final reported model relies on predictors supported by both RFE and embedded feature importance methods.

Model training and validation

Eight ML algorithms were implemented, including Random Forest (RF), Support Vector Machine (SVM), Gradient Boosting Machine (GBM), Neural Network (NNET), eXtreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), k-Nearest Neighbors (KNN), and Adaptive Boosting (AdaBoost). To reduce the risk of overfitting, all models were trained using

tenfold stratified cross-validation, with hyperparameters optimized via the caret resampling framework (Supplementary Table 1). Model discrimination was assessed in the internal cross-validation folds and in an independent external validation cohort. Calibration was evaluated using calibration plots (reliability curves) and the Hosmer–Lemeshow goodness-of-fit test in both cohorts. Subsequently, based on these final hyperparameters and the best feature subset, the models were fitted on the training set. Model evaluation metrics included receiver operating characteristic (ROC), recall, precision, F1 score, accuracy, Brier score, and the area under curve (AUC), with corresponding calculation formulas detailed in Supplementary Table 2. Decision curve analysis (DCA) was further applied to evaluate the clinical net benefit of the model across a range of threshold probabilities.

SHAP interpretable framework

The SHapley Additive exPlanations (SHAP) approach was applied to enhance the transparency of ML model predictions by assigning quantitative contributions to each input variable for every individual prediction. Compared with traditional feature importance methods, SHAP not only identifies the relative importance of predictive variables, but also reveals the directionality (positive or negative) of their association with the outcome of interest. Additionally, it supports both global interpretability (feature impact across the dataset) and local interpretability (feature contributions for individual predictions). In this study, SHAP was utilized to explore and visualize the most influential features associated with MACE in T2DM patients, offering insight into both population-level and personalized risk attribution.

Statistical analysis

Statistical analyses were conducted using IBM SPSS Statistics 26.0. Continuous variables were summarized as mean ± standard deviation (SD) or median with interquartile range (IQR) based on distribution, and group differences were assessed using Student's t-test or Mann–Whitney U test as appropriate. Categorical data were presented as counts (%) and compared using Chi-square test or Fisher's exact test. To identify candidate predictors of MACE, univariate logistic regression analysis was first conducted in the training cohort. Variables with a *P*-value < 0.05 were retained for further analysis. A two-sided *p*-value < 0.05 was regarded as statistically significant.

All ML algorithms and SHAP interpretability analyses were conducted in R 4.4.2. For ML model development and evaluation, R packages including *tidymodels*, *caret, randomForest*, *glmnet*, *XGBoost*, *rpart*, *gbm*, *neuralnet*, *e1071*, *kernlab*, and *pROC*, alongside *doParallel* and *MLmetrics* for parallel computing and performance metrics. Model interpretability was implemented using the *DALEX* and *shapviz* packages. Data visualization and result presentation were performed using *ggplot2*, *ggbeeswarm*, and *forcats*. Additionally, the *VIM* and *Boruta* packages were employed for data preprocessing and feature selection.

Results

Baseline characteristics

A total of 686 patients with T2DM were included in the study, of whom 59% were male, with a mean age of 62.9 ± 10.82 years. During the follow-up period, 183 patients (27%) experienced MACE. Compared with those who did not experience MACE, patients in the MACE group exhibited significantly higher BMI (26.56 ± 3.56 vs. 24.66 ± 3.31 kg/m2, P < 0.001) and elevated PCAT-FAI values (RCA-FAI: − 71.85 ± 12.72 vs. − 81.61 ± 13.03 HU, LAD-FAI: 79.46 ± 11.46 vs. − 82.88 ± 11.42 HU, LCX-FAI: − 73.00 ± 11.73 vs. − 77.31 ± 11.63 HU, P < 0.001) (Table 1). Supplementary Table 3 presents a comparison of the training and test datasets. Statistical analysis shows that the distributions of most variables are comparable between the two datasets, and the differences are not statistically significant (most P values > 0.05).

Feature selection

Univariate logistic regression was applied to the training dataset to identify variables significantly associated with MACE. A total of 11 variables with $P < 0.05$ were identified, including gender, BMI, HDL-C, eGFR, MONO, MHR, significant stenosis, RCA-FAI, LAD-FAI, LCX-FAI, RCA volume index (Supplementary Table 4).

Subsequently, the RFE algorithm nested in the SVM model was used to further screen the above variables. Based on the model's performance, eight variables were retained as the optimal subset of predictors: BMI, MHR, HDL-C, eGFR, MONO, RCA-FAI, LCX-FAI, and RCA volume index (Fig. 3).

Model development and performance evaluation

A total of eight ML models were developed and optimized through hyperparameter tuning. Each model was subsequently trained on the entire training set and evaluated on an independent test set to assess predictive performance (Fig. 4). Among all algorithms, the XGBoost model demonstrated the most robust and consistent performance. In the training cohort, it achieved a high discriminative ability with an AUC of 0.818 (95% *CI* 0.777–0.858). Although Random Forest and LightGBM achieved perfect discrimination (AUC = 1.000) in the training cohort, their external performance declined and calibration was poor. By contrast, the XGBoost model

Table 1 Baseline analysis result of 686 patients with and without MACE in T2DM

	All n = 686	MACE n = 183	No-MACE n = 503	*P*-value
Men, n (%)	403(59)	121(66)	282(56)	0.017
Age (y)	62.91 ± 10.82	62.17 ± 11.82	63.17 ± 10.44	0.284
BMI (kg/m^2)	25.16 ± 3.48	26.56 ± 3.56	24.66 ± 3.31	< 0.001
SBP (mmHg)	137.88 ± 20.03	138.32 ± 20.19	137.72 ± 19.99	0.728
DBP (mmHg)	80.02 ± 12.43	80.91 ± 13.06	79.70 ± 12.19	0.259
Smoking, n(%)	211(31)	58 (32)	153 (30)	0.748
Hypertension, n (%)	480(70)	132 (72)	348 (69)	0.456
Dyslipidemia, n (%)	378(55)	100 (55)	278 (55)	0.885
Hyperuricemia, n (%)	221(32)	61 (33)	160 (32)	0.705
Statin, n (%)	385(56)	109 (60)	276 (55)	0.273
β-Blocker, n (%)	177(26)	51 (28)	126 (25)	0.455
ACEI or ARB, n (%)	225(33)	67 (37)	158 (31)	0.199
CCB, n (%)	223(33)	64 (35)	159 (32)	0.405
HbA1c (%)	7.46 ± 1.60	7.56 ± 1.71	7.43 ± 1.56	0.34
TG (mmol/L)	2.09 ± 1.86	2.23 ± 1.80	2.04 ± 1.88	0.22
TC (mmol/L)	4.64 ± 1.22	4.61 ± 1.15	4.65 ± 1.24	0.674
HDL-C (mmol/L)	1.06 ± 0.28	0.97 ± 0.24	1.09 ± 0.29	< 0.001
LDL-C (mmol/L)	2.91 ± 1.04	2.85 ± 0.98	2.94 ± 1.07	0.358
eGFR (mL/min/1.73m2)	82.34 ± 22.78	78.66 ± 24.23	83.68 ± 22.10	0.01
MONO (× 10*9/L)	0.508 ± 0.181	0.557 ± 0.220	0.490 ± 0.161	< 0.001
MHR	0.519 ± 0.256	0.613 ± 0.302	0.485 ± 0.228	< 0.001
Significant stenosis, n (%)	293(43)	91 (49.7)	202 (40)	0.025
CAD-RADS category				0.065
CAD-RADS 0, n (%)	161(24)	40(22)	121(24)	
CAD-RADS 1, n (%)	97(14)	20(11)	77(15)	
CAD-RADS 2, n (%)	135(20)	32(17)	103(20)	
CAD-RADS 3, n (%)	115(17)	29(16)	86(17)	
CAD-RADS 4, n (%)	151(22)	50(27)	101(20)	
CAD-RADS 5, n (%)	27(4)	12(7)	15(3)	
RCA-FAI(HU)	− 79.01 ± 13.64	− 71.85 ± 12.72	− 81.61 ± 13.03	< 0.001
LAD-FAI(HU)	− 81.97 ± 11.52	− 79.46 ± 11.46	− 82.88 ± 11.42	< 0.001
LCX-FAI(HU)	− 76.16 ± 11.81	− 73.00 ± 11.73	− 77.31 ± 11.63	< 0.001
RCA volume index (cm^3)	1.89 ± 0.55	1.80 ± 0.57	1.92 ± 0.54	0.012
LAD volume index (cm^3)	1.83 ± 0.50	1.79 ± 0.52	1.84 ± 0.49	0.273
LCX volume index (cm^3)	1.27 ± 0.47	1.21 ± 0.49	1.29 ± 0.46	0.037

BMI body mass index, *SBP* systolic blood pressure, *DBP* diastolic blood pressure, *ACEI* angiotensin-converting enzyme inhibitor, *ARB* angiotensin-receptor blocker, *CCB* calcium channel blockers, *HbA1c* glycated hemoglobin A1c, *TG* triglycerides, *TC* total cholesterol, *HDL-C* high-density lipoprotein cholesterol, *LDL-C* low-density lipoprotein cholesterol, *eGFR* estimated glomerular filtration rate, *MONO* monocyte count, *MHR* MONO/HDL-C, *MACE* major adverse cardiac events; *CAD-RADS* Coronary Artery Disease – Reporting and Data System, *RCA-FAI* right coronary artery-fat attenuation index, *LAD* left anterior descending artery, *LCX* left circumflex artery

maintained good discrimination (AUC = 0.809, 95% CI 0.700–0.918) and acceptable calibration (Hosmer–Lemeshow $\chi^2[8] = 6.50$, $p = 0.592$), underscoring its robustness and clinical applicability (Supplementary Fig. 1). Relative to the clinical model, adding PCAT-FAI yielded significant incremental value in the external cohort—categorical NRI = 0.1794 (95% CI 0.1081–0.2506, $p < 0.001$), continuous NRI = 0.4864 (95% CI 0.3114–0.6615, $p < 0.001$), and IDI = 0.0692 (95% CI 0.0477–0.0907, $p < 0.001$). DCA further showed the highest net benefit across a wide range of threshold probabilities in both the training and validation sets.

To provide a comprehensive assessment of model performance, additional metrics were calculated, including recall, precision, accuracy, F1 score, and Brier score (Table 2). The XGBoost model showed balanced

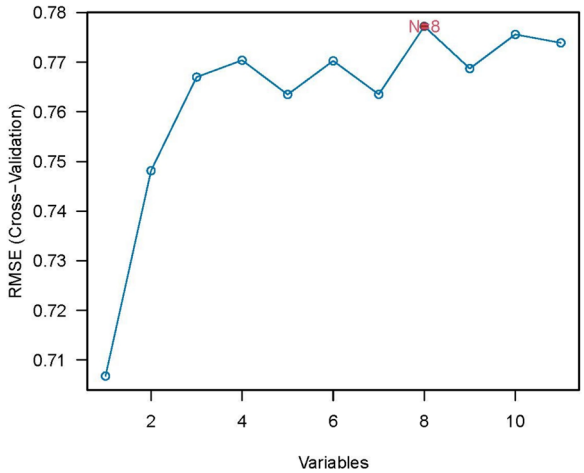


Fig. 3 The RFE algorithm based on the SVM model further screens variables

performance with sensitivity of 0.654, specificity of 0.882, accuracy of 0.824, F1 score of 0.654, and Brier score of 0.248. These findings underscore the potential of XGBoost as a practical and effective tool for long-term cardiovascular risk prediction in patients with T2DM.

Feature importance ranking

To further evaluate the relative importance of predictive variables, feature importance was evaluated across all eight machine learning models (Fig. 5). Among all models, RCA-FAI consistently ranked as the top predictor in all models, indicating that its strong and stable association with long-term MACE risk among patients with T2DM.

Other features that frequently appeared among the top-ranking variables included BMI, MHR, and HDL-C, indicating the relevance of obesity, systemic inflammation, and lipid metabolism in cardiovascular risk stratification.

SHAP analysis

To improve the interpretability of the XGBoost model, SHAP analysis was conducted to assess the individual contribution of each input feature to the model's predictions. As shown in Fig. 6A, the mean absolute value of SHAP indicates that RCA-FAI is the most influential feature, followed by BMI, MHR, HDL-C, and MONO. This ranking was consistent with the feature importance scores derived by the model. The SHAP summary plot (Fig. 6B) further illustrated the direction and magnitude of feature effects. Specifically, lower RCA-FAI values and higher BMI and MHR were associated with an increased risk of MACE, underscoring their pivotal role in risk stratification for patients with T2DM.

To further illustrate feature contributions at the individual prediction level, SHAP force plots were generated for representative high-risk patients (Fig. 6C and D). The results showed that low RCA-FAI (−63), elevated BMI (26.7), and increased MHR (0.52) cumulatively contributed positively to the model output, resulting in a high predicted probability of MACE. Conversely, variables such as higher HDL-C and eGFR showed negative SHAP values, indicating potential protective effects. Overall, SHAP analysis confirmed the importance and clinical relevance of RCA-FAI, BMI, and MHR in predicting cardiovascular risk and provided a clear framework for individualized risk interpretation.

Discussion

In this study, we systematically compared the performance of eight ML algorithms in a T2DM follow-up cohort of 686 patients, identified the predictive role of PCAT parameters in MACE risk. Incorporating PCAT parameters into the predictive framework, particularly in the XGBoost model, resulted in excellent performance on the test set (AUC = 0.809). Among all features, RCA-FAI consistently emerged as the most influential predictor, outperforming traditional clinical variables in importance. This was further validated by Shapley analysis, reinforcing its strong association with MACE. These findings underscore the potential of PCAT imaging biomarkers as novel and individualized tools for cardiovascular risk assessment in patients with T2DM.

Previous studies have explored the prediction of MACE in patients with T2DM using conventional statistical approaches, such as logistic regression models [21]. While these models often fall short in capturing complex nonlinear interactions among variables, which limits their predictive performance. To address this limitation, this study used the RFE strategy for variable selection based on traditional univariate analysis, systematically eliminating redundant features, thereby improving the robustness and accuracy of the prediction model. This approach facilitated the selection of features with the highest predictive contribution, while simultaneously minimizing model complexity and reducing the likelihood of overfitting [26]. The RFE strategy has been shown to be effective in risk stratification of various chronic diseases [27].

Among the 8 ML models evaluated in this study, the XGBoost model performed best, with the highest AUC value, and showed good performance in terms of accuracy, calibration, and net clinical benefit. In the external validation set, the XGBoost model also maintained superior predictive accuracy, indicating strong generalizability. As a powerful ensemble learning algorithm, XGBoost is capable of handling high-dimensional, nonlinear, and

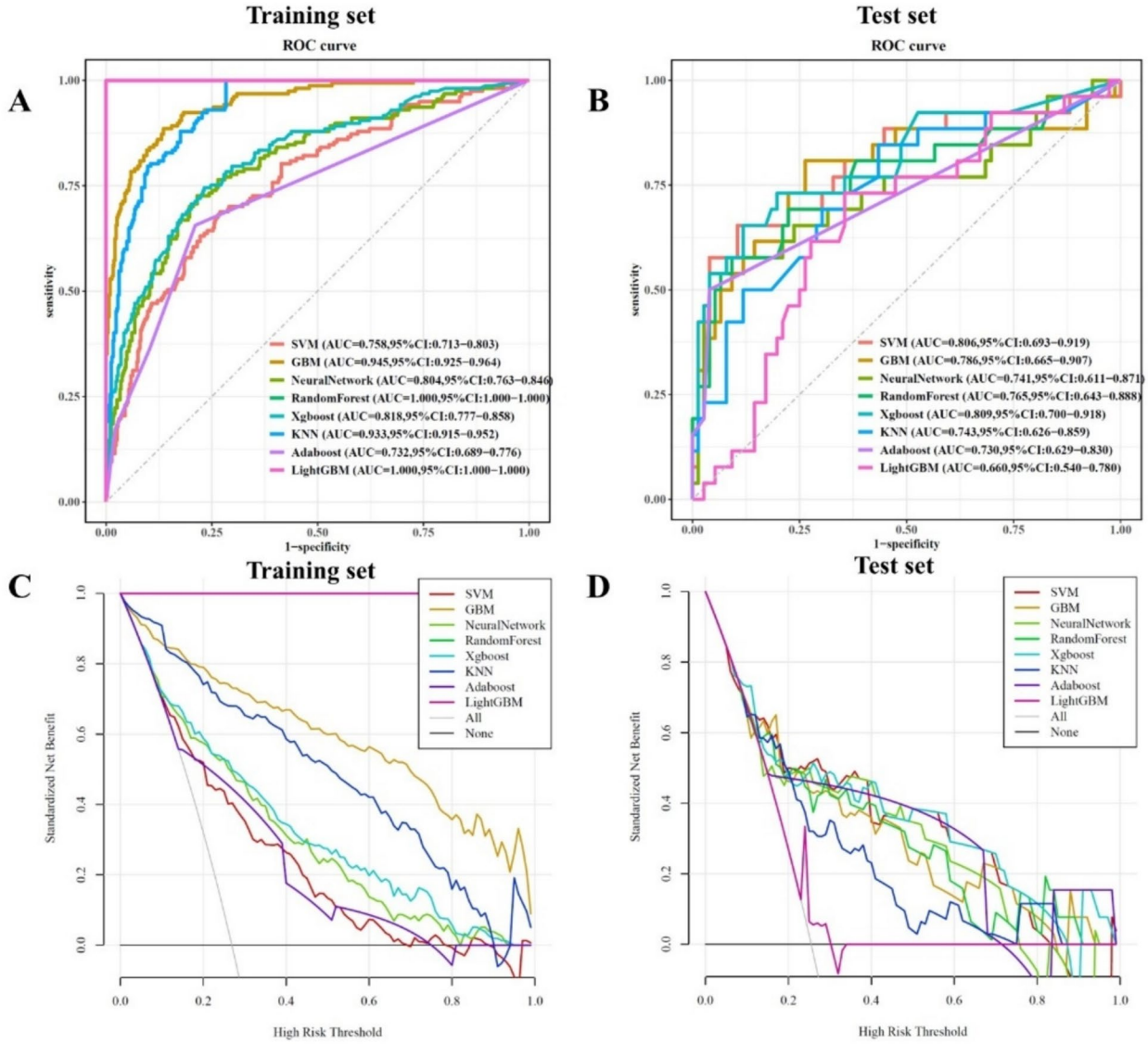


Fig. 4 Performance comparison of different ML models in the training and test sets. **A** and **C** Are ROC curve analyses and DCA curves in the training set. **B** and **D** Are ROC curve analyses and DCA curves in the test set. *SVM* Support Vector Machine, *GBM* Gradient Boosting Machine, *NNET* Neural Network, *RF* Random Forest, *XGBoost* eXtreme Gradient Boosting, *KNN* k-Nearest Neighbors, *AdaBoost* Adaptive Boosting, *LightGBM* Light Gradient Boosting Machine

Table 2 Predictive performance comparison of eight ML models

Model	Threshold	Accuracy	Sensitivity	Specificity	Precision	F1	Brier score
SVM	0.239	0.833	0.654	0.895	0.680	0.667	0.157
GBM	0.126	0.755	0.808	0.737	0.512	0.627	0.145
NNET	0.305	0.853	0.538	0.961	0.824	0.651	0.145
RF	0.238	0.824	0.577	0.908	0.682	0.625	0.149
XGBoost	0.498	0.824	0.654	0.882	0.654	0.654	0.248
KNN	0.138	0.637	0.846	0.566	0.400	0.543	0.161
AdaBoost	0.212	0.843	0.500	0.961	0.812	0.619	0.163
LightGBM	0.008	0.667	0.731	0.645	0.413	0.528	0.288

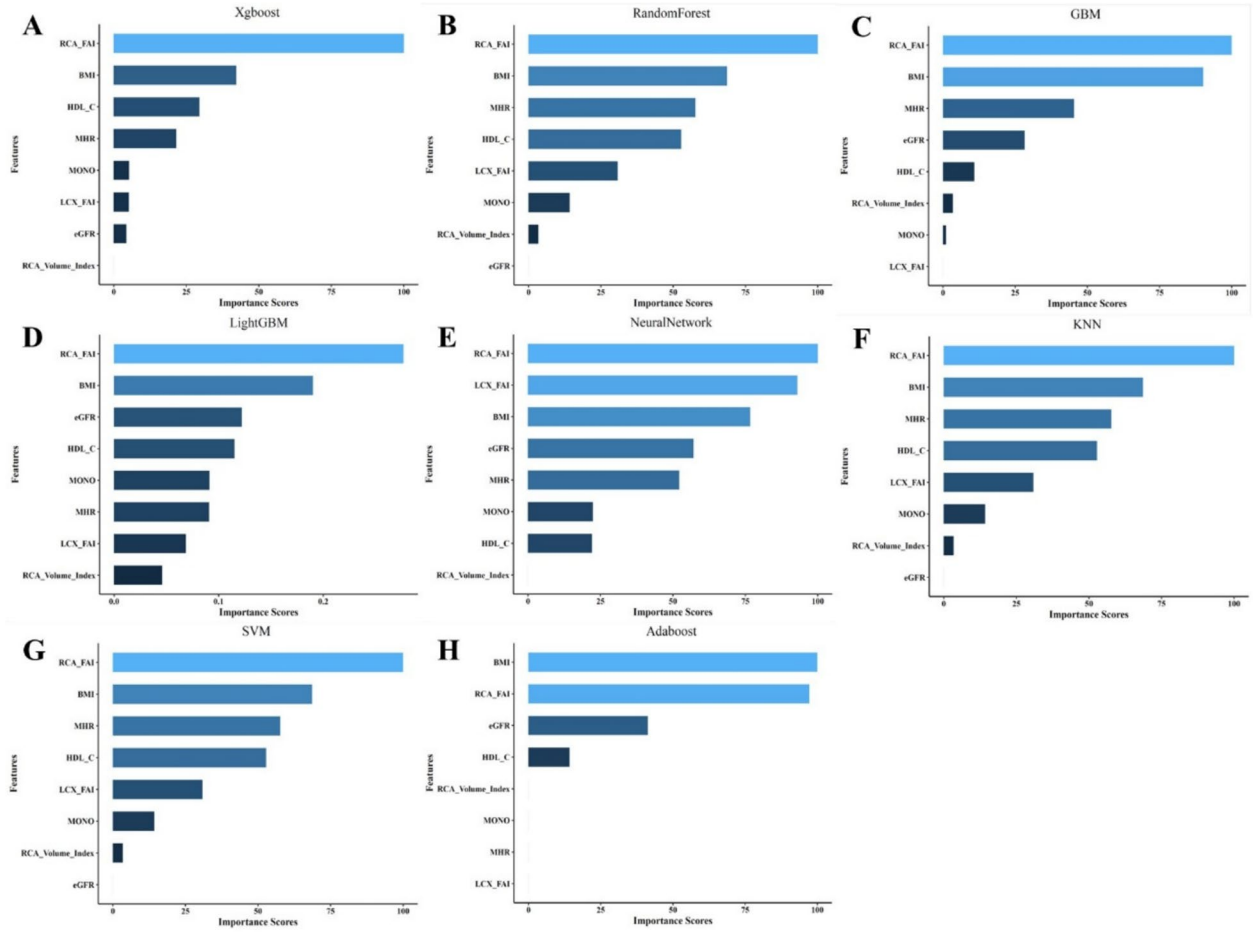


Fig. 5 Feature importance ranking for each machine learning model. A–H Correspond to XGBoost, RF, GBM, LightGBM, NNET, KNN, SVM, and AdaBoost, respectively

heterogeneous clinical data, and has been widely used in medical prediction research in recent years. Based on this method, we constructed a final prediction model containing 8 features. Therefore, this model is expected to become an effective tool for predicting the long-term risk of MACE in T2DM patients, providing support for precise risk stratification and individualized intervention strategies [28, 29].

Among all input variables, RCA-FAI consistently ranked first in all machine learning models, both in feature importance ranking and SHAP analysis. This finding highlights the potential of RCA-FAI as a stable imaging biomarker for cardiovascular risk assessment in patients with T2DM. Mechanistically, high PCAT attenuation reflects a pro-inflammatory adipose tissue phenotype, characterized by the paracrine release of cytokines and chemokines [30]. These inflammatory mediators are implicated in promoting necrotic lipid core formation and thinning of fibrous caps—hallmarks of plaque vulnerability [31, 32]. In T2DM, macrophage polarization towards a pro-inflammatory M1 phenotype, which is driven by insulin resistance, may further intensify these processes [33–35]. Anatomically, the RCA is surrounded by a relatively abundant volume of adipose tissue and is less affected by adjacent large-caliber vessels, thereby offering a clearer representation of coronary inflammatory status and systemic cardiovascular risk. This anatomical and pathophysiological context likely accounts for the superior predictive value of RCA-FAI compared to LAD-FAI or LCX-FAI [23]. The results of this study are consistent with previous studies, which also found that higher PCAT fat attenuation was associated with adverse cardiovascular outcomes, although most of these studies were conducted in non-diabetic populations [36–39]. Given the chronic inflammatory state and metabolic disorders in patients with T2DM, the predictive value of RCA-FAI may be more prominent. Our study further pointed out that RCA-FAI outperformed traditional risk factors in the machine learning model, showing its additional predictive value. Although direct comparison

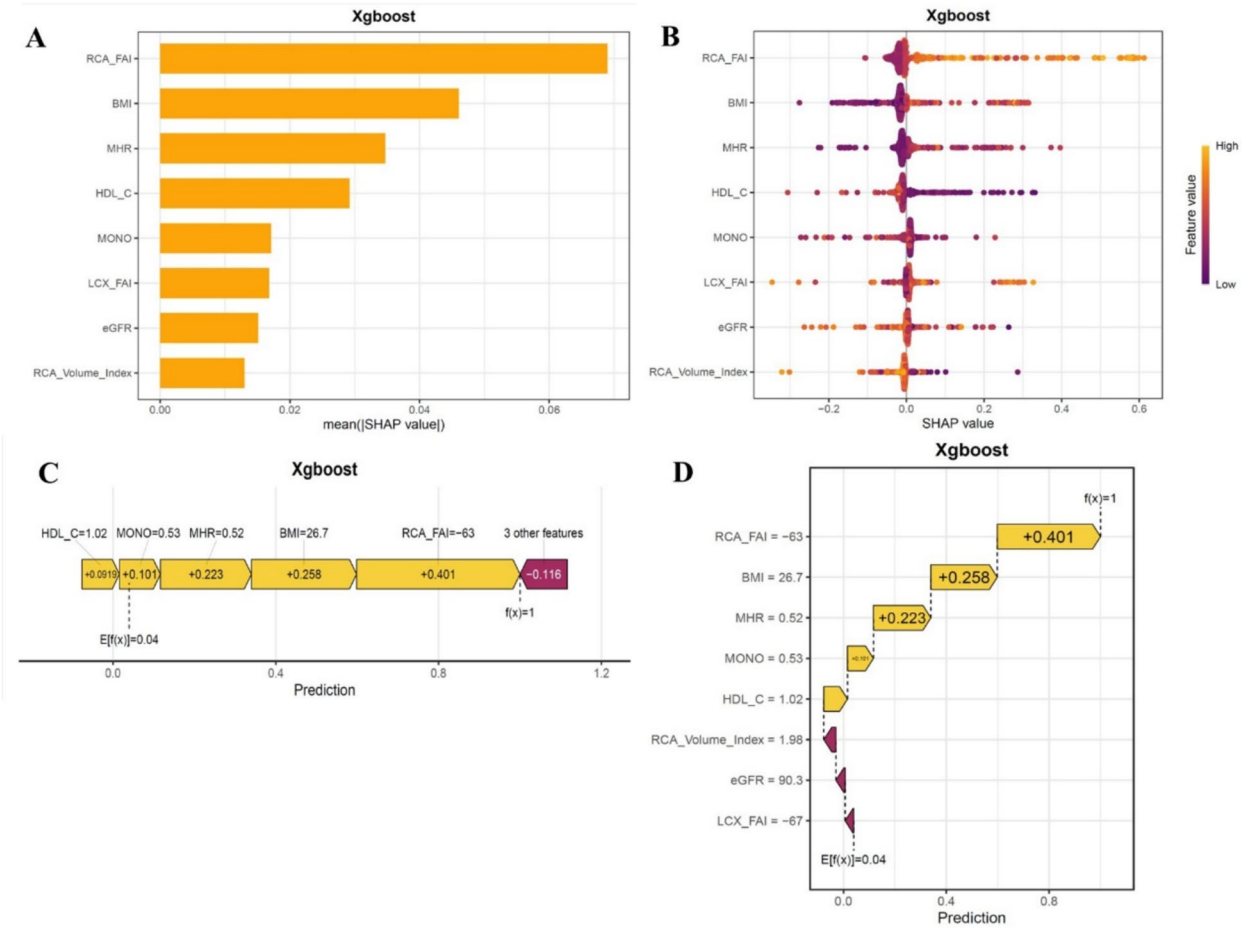


Fig. 6 SHAP analysis for the XGBoost model. A Mean absolute SHAP values for each feature, indicating overall contribution to the model. B SHAP summary plot showing the distribution and direction of SHAP values for each feature. C SHAP force plot for an individual case with high predicted MACE risk. D SHAP decision plot showing the cumulative effect of each feature on the final prediction

with widely used cardiovascular risk scores such as the Framingham or ESC/EASD models was not feasible in our cohort due to incomplete alignment of required variables [8], we recognize that such analyses would further contextualize the added value of PCAT-FAI. Nonetheless, the incremental performance metrics presented in this study provide strong evidence that incorporating PCAT-FAI meaningfully improves risk prediction beyond conventional clinical factors. Future multicenter studies with broader datasets will enable direct benchmarking against established risk scores.

The XGBoost prediction model constructed in this study finally included 8 key variables, including RCA-FAI, BMI, MHR, HDL-C, eGFR, MONO, RCA volume index, and LCX-FAI. These features were screened by the RFE strategy and showed strong contributions in the SHAP analysis. Among these, BMI and MHR reflect the systemic metabolic and inflammatory state. High BMI, a typical manifestation of metabolic disorders, has been widely associated with increased MACE risk in patients with T2DM. Obesity represents not merely excess weight but a complex pathological condition characterized by insulin resistance, hyperglycemia, hypertension, and dyslipidemia, all of which are established contributors to cardiovascular risk [40]. MHR, a novel inflammatory biomarker derived from the ratio of monocyte count to HDL-C, has recently emerged as a prognostic indicator in diabetic, obese, and acute coronary syndrome populations [41, 42]. Our results further validate the predictive value of MHR in the T2DM population. Consistent with previous studies, HDL-C levels in our cohort were negatively associated with MACE risk. This observation is supported by a large-scale retrospective study of approximately 600,000 T2DM patients, which revealed a U-shaped correlation between HDL-C levels and both cardiovascular morbidity and all-cause mortality, indicating that deviations from optimal HDL-C ranges—whether too low or too high—are linked to

worse outcomes [43]. eGFR reflects the state of renal function and is a key indicator in the cross-pathology of chronic kidney disease and cardiovascular disease. Previous research has established that decreased eGFR is independently correlated with increased cardiovascular and all-cause mortality [44]. This suggests that impaired renal function may portend elevated MACE risk. Finally, MONO, reflecting absolute monocyte count, serves as an indirect indicator of chronic inflammation and immune activation. Increasing evidence has shown that monocytes actively contribute to the initiation, progression, and destabilization of atherosclerotic plaques, particularly in the context of immune dysregulation seen in diabetes [45].

Beyond statistical performance, the model's output can be translated into pragmatic steps at the point of care. First, model-estimated risk may be used to categorize patients into low/intermediate/high risk (using prespecified thresholds or decision-curve–informed quantiles) to guide the intensity of follow-up. Second, patients who fall into a high-risk stratum and/or display adverse profiles on key predictors (RCA-FAI, BMI, MHR) may be prioritized for closer surveillance (e.g., shorter clinical review intervals) and more stringent optimization of guideline-directed cardiometabolic therapy; functional testing may be considered when clinically indicated. Third, in resource-limited settings without ready access to CCTA, MHR—obtainable from routine laboratory tests—could serve as a low-cost adjunct to clinical assessment for triage and longitudinal monitoring. These applications are intended to complement, not replace, prevailing clinical guidelines and should be locally calibrated, contingent on standardized acquisition/analysis and further validation in broader cohorts.

This study has several limitations. First, it is a single-center retrospective analysis and, although external validation used data from a branch hospital within the same health system, broader transportability across scanners, institutions, and calendar time remains uncertain; multicenter and temporally distinct datasets are needed. At present, our study provides a transparent modeling framework and openly described analysis pipeline, thereby facilitating replication and extension by independent investigators in diverse settings. Second, the generalizability of PCAT-FAI may be constrained by technical variability (scanner/vendor, protocol, reconstruction, post-processing). Although we applied a uniform acquisition protocol within our health system, reproducibility across institutions remains uncertain. Future work should prioritize standardized acquisition/analysis, consensus guidance, and harmonization (e.g., ComBat) to ensure methodological consistency. Third, as an initial dimensionality-reduction step, variables with significant associations in univariate logistic regression were retained for further analysis. While practical, this approach may introduce bias and risk excluding predictors that exert influence only through multivariable or nonlinear interactions. Future studies with larger cohorts should consider embedded feature-selection strategies that avoid reliance on univariate pre-filtering. In addition, missing data were imputed with MICE (m = 5) without formal convergence diagnostics, which may marginally increase uncertainty. Fourth, the external cohort was modest in size with few MACE events, limiting the precision of discrimination, calibration, and net-benefit estimates. Larger and more diverse validation datasets are needed to confirm the robustness of our findings. Although this study modeled MACE as a binary outcome over an average follow-up of 38.4 months, thereby facilitating algorithmic comparisons, this design does not fully account for variation in follow-up duration. Future studies should extend the framework into survival-analysis models that incorporate censoring and generate time-specific risk predictions, which would provide additional clinical utility. Finally, future work should elucidate the biological mechanisms linking PCAT parameters to MACE in T2DM and validate the model in larger, more diverse populations, thereby enhancing both the accuracy and the applicability of cardiovascular risk prediction tools.

Conclusions

In this study, we developed a machine learning model to predict MACE risk in T2DM patients based on clinical data extracted from four branches of a hospital. The XGBoost model demonstrated robust performance in both internal and external validation. These findings provide an evidence-based tool that may assist clinicians in precisely identifying high-risk individuals with T2DM and facilitate the development of personalized preventive strategies.

Abbreviations

PCAT	Pericoronary adipose tissue
T2DM	Type 2 diabetes mellitus
MACEs	Major adverse cardiovascular events
SHAP	SHapley Additive exPlanations
ML	Machine learning
BMI	Body mass index
IDF	International diabetes federation
CVD	Cardiovascular disease
AHA	American heart association
LDL-C	Low-density lipoprotein cholesterol
CCTA	Coronary computed tomography angiography
FAI	Fat attenuation index
CAD	Coronary artery disease
ADA	American diabetes association
PCI	Percutaneous coronary intervention
CABG	Coronary artery bypass grafting
RCA	Right coronary artery
LAD	Left anterior descending artery

LCX	Left circumflex artery
HU	Hounsfield units
HDL-C	High-density lipoprotein cholesterol
TG	Triglyceride
TC	Total cholesterol
MONO	Monocyte count
HbA1c	Glycosylated hemoglobin
eGFR	Estimated glomerular filtration rate
TG	Triglyceride
TC	Cholesterol
SBP	Systolic blood pressure
DBP	Diastolic blood pressure
MHR	Monocyte to high-density lipoprotein ratio
DCA	Decision curve analysis
RFE	Recursive feature elimination
RF	Random forest
SVM	Support vector machine
XGBoost	EXtreme gradient boosting
GBM	Gradient boosting machin
KNN	K-nearest neighbors
NNET	Neural network
AdaBoost	Adaptive boosting
LightGBM	Light gradient boosting machine
ROC	Receiver operating characteristic
AUC	Area under curve

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40001-025-03237-4.

Supplementary file 1

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Author contributions
Y.T. and X.Z. formulated and designed the research, conducted data analysis and interpretation. X.Y., S.G. and Q.L. were responsible for data collection and patient follow-up. M.S. and H.C. drafted the manuscript. W.Z. and H.W. performed data analysis. Y.L. and G.L. critically revised the manuscript. L.W. supervised the entire study and provided overall guidance. All authors interpreted the results and approved the final manuscript for submission.

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Data availability
The datasets used during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
The study data collection was approved by the Ethics Committee of Guangdong Provincial Hospital of Traditional Chinese Medicine. Written informed consent from participants was not required because of the study's retrospective design and the use of anonymized data, in compliance with national and institutional regulations.

Competing interests
The authors declare no competing interests.

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